

Rongrong Liu

Natural Language Processing · Retrieval-Augmented Generation · Multimodal Learning · LLM Evaluation

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Education

Nanjing University & Johns Hopkins University Sep 2025 – Present
M.A. Program in International Studies (International Economics) — Hopkins–Nanjing Center, SAIS GPA: 3.85/4.0
First-Class Graduate Scholarship; Nanjing Scholar; fully funded study and research year at Johns Hopkins University, Washington, DC.

University of California, Berkeley Jan 2024 – May 2024
Visiting Student, Haas School of Business GPA: 4.0/4.0

Xi'an Jiaotong University Sep 2021 – Jun 2025
B.A. in Finance Rank: 11/75 (Top 15%), GPA: 3.49/4.3
Coursework: Advanced Mathematics, Linear Algebra, Probability & Statistics, C/C++, Python.

Research Experience

Multimodal Generative Models for Controllable Video Editing Sep 2025 – Present
School of Intelligent Science and Technology, Nanjing University — Prof. Zili Yi's Lab

- Co-authored “**InstructVVT: Instruction-Driven Video Virtual Try-On without Auxiliary Spatial Priors,**” under review at *NeurIPS 2026*; the framework combines a Diffusion Transformer (DiT), MLLM-derived semantic guidance, reference-garment conditioning, and language-based target specification without inference-time spatial priors.
- Built automated data pipelines generating and filtering **1,000+ multimodal video-editing samples**; developed evaluation protocols for instruction alignment, appearance fidelity, structural preservation, and temporal consistency.
- Surveyed and reproduced representative controllable video-editing approaches and organized major editing settings into a technical taxonomy to support benchmark design and comparative evaluation.

Faithfulness in Knowledge-Intensive Retrieval-Augmented Generation Nov 2025 – Present
Independent Research; collaboration with researchers from UIUC and the First Affiliated Hospital of Zhengzhou University

- Redesigned HippoRAG-2-style knowledge organization around relations and logical structures and built experimental pipelines for compositional, cross-document **multi-hop retrieval and reasoning**.
- Extended the work to citation-faithful clinical RAG, designing evaluation for source correctness, locator-level attribution, and abstention; an expert-validated pilot exposed citation-location errors and failures on unanswerable questions.

AI & Research Engineering Experience

VectifyAI (PageIndex) — AI Backend / Algorithm Intern Dec 2025 – Mar 2026

- Built and integrated Mistral/Datalab OCR Server backends supporting the product's **Reference to Lines** functionality; additionally integrated Azure Document Intelligence and Google Document AI, normalizing heterogeneous OCR outputs into a unified **block-level representation** for downstream retrieval.
- Helped refactor page-level indexing into **block-level semantic retrieval** for RAG document analysis; standardized authentication and configuration across document-intelligence backends and built reproducible backend environments across development and testing.
- Wrote technical handoff and deployment documentation and supported intern onboarding, translating system architecture, configuration, and testing workflows into reproducible engineering procedures.

Granite Asia (formerly GGV Capital) — Research Intern, Embodied AI Mar 2026 – Jun 2026

- Built a multi-axis framework for analyzing **20+ frontier embodied-AI systems** across architecture, training data, controllability, latency, and closed-loop readiness, covering world models, robot learning, and vision-language-action models.
- Synthesized frontier research and **20+ expert interviews / technical due-diligence discussions** with AI researchers and robotics startups into technical landscape analyses of emerging model paradigms.

Shenwan Hongyuan Securities, FICC — LLM Engineering Intern Jul 2025 – Aug 2025

- Deployed Qwen-family LLMs on CentOS/Ubuntu servers and built automated evaluation and monitoring pipelines for financial-news analysis, reducing the system's missed-detection rate from **3% to 0.1%**.

Selected Project

Global AI Research & Capability Trend Dashboard Nov 2025
ai-trend-478401.web.app

- Built and deployed an end-to-end research analytics platform covering **30+ AI subfields (2010–2025)**, with an OpenAlex–GCP data pipeline, Python backend, React/Vite analytics interface, Firebase deployment, automated tests, and Docker support.

Open-Source Contribution

HippoRAG (NeurIPS 2024) — Open-Source Contributor 2026

- Identified a Python/OpenAI dependency compatibility issue in the local development setup and proposed a fix via **PR #176**; the maintainers subsequently updated the dependency configuration.

Technical Skills

Programming: Python, C/C++ **AI/ML:** LLMs, RAG, Information Retrieval, Multimodal Models, Diffusion Models, OCR/Document AI

Research Methods: Benchmark Design, Error Analysis, LLM Evaluation, Conformal Prediction, Statistical Analysis

Systems: Docker, Git, Linux, GPU Servers, REST APIs, GCP/Firebase **Languages:** English (IELTS 7.5)